Practical No. 10

Name : Abhay Ramesh Shinde

Class : SY-B

Branch – AI & DS

Roll no : B-44

Aim - Version Control using Git & GitHub Questions :

1. **What is the difference between Git and GitHub? Ans**

: Git vs GitHub Git:

• Git is a Version Control System (VCS)

• Created by Linus Torvalds in 2005

• Works completely offline on your local machine

• Tracks every change made to your code

• Stores full project history locally

• Free and open-source

• Used via command line (terminal)

• Alternatives: Mercurial, SVN GitHub:

• GitHub is a cloud-based hosting platform for Git repos

• Founded in 2008, bought by Microsoft in 2018

• Requires internet connection

• Adds collaboration features: Pull Requests, Issues, Actions

• Stores your code remotely on servers

• Has free and paid plans

• Alternatives: GitLab, Bitbucket

2. Explain the difference between:

- git add

- git commit

Ans : git add:

• Moves files from Working Directory → Staging Area

• Tells Git which changes to include in next commit

• Does NOT save permanently

• Can stage one file, multiple files, or all files

• Reversible using git restore –staged

Commands:

Git add file.txt → one file

Git add . → all files

Git add *.js → all JS files

Git add -p → select parts interactively Git

commit:

• Moves files from Staging Area → Local Repository

• Creates a permanent snapshot in history

• Each commit gets a unique SHA-1 hash ID

• Requires a commit message • Records author, timestamp, and

changes Commands:

Git commit -m “message” → basic commit

Git commit –amend → edit last commit Git commit -

am “message” → add + commit together

3.What is the staging area in Git?

Ans : • Also called the Index

• It is a middle layer between working directory and repository

• Physically stored in .git/index file

• Lets you selectively choose what to commit

• You can modify 10 files but commit only 2

• Helps create clean, logical commits • You can review staged

changes before saving

- Commands:

Git status → see staged/unstaged files

Git diff –staged → see staged changes Git

restore –staged file → unstage a file -

4 States of a file:

• Untracked → new file Git doesn't know about

• Modified → tracked file with new changes

• Staged → ready to be committed

• Committed → permanently saved in history

4**. What happens if you modify a file after committing it?**

Ans : • The old committed version stays safe in history

• Modified file appears as "modified" in working directory

• Git detects change by comparing file hash with last commit

• Git status shows it under "Changes not staged for commit"

• You must add and commit again to save new version

• Each commit gets a new unique hash

• Old commit is still accessible anytime

• You can go back using git checkout <old-hash>

• Nothing in Git is truly lost after committing

• To fix last commit quickly: git commit –amend Process after modifying:

Modify file → git add → git commit → new snapshot created

5.Difference between:

- git push

- git pull

Ans : git push:

• Uploads local commits → remote repository (GitHub)

• Shares your work with teammates

• Only pushes committed changes (not just staged)

• May be rejected if remote has newer commits Commands:

- Git push origin main → push to main branch
- Git push origin branch-name → push specific branch
- Git push -u origin main → set default upstream
- Git push –force → force overwrite (dangerous) Git

pull:

• Downloads remote changes → local machine

• Keeps your local repo up to date

• Combines: git fetch + git merge

• Can cause merge conflicts if same lines were edited Commands:

- Git pull origin main → pull from main

-Git pull –rebase → pull and rebase instead of merge

6 .What is Version Control? Differentiate between Centralized v/s Distributed VCS.

Ans : Version Control:

• System that records changes to files over time

• Tracks who, what, when, and why something changed

• Allows reverting to any previous version

• Enables parallel work without overwriting each other

• Essential for team software development Centralized VCS

(CVCS):

• One central server holds the entire repository

• Developers check out only one version (not full history)

• All operations need server/internet connection

• Examples: SVN, CVS, Perforce - Advantages:

• Simple to understand

• Admin has full control

• Good for binary/design files Distributed VCS (DVCS):

• Every developer has full copy of entire repo including history

- Most operations happen locally (fast)
- Remote server used only for syncing/sharing
- Examples: Git, Mercurial - Advantages:
- Fully offline capability
- Every clone = complete backup
- Very fast operations
- Easy branching and merging• No single point of failure **Theroy :**

```
enovo@LAPTOP-80NSFJM5 MINGW64 ~/Myproject (master)
$ mkdir myproject

enovo@LAPTOP-80NSFJM5 MINGW64 ~/Myproject (master)
$ cd myproject

enovo@LAPTOP-80NSFJM5 MINGW64 ~/Myproject/myproject (master)
$ git init
Initialized empty Git repository in C:/Users/Lenovo/Myproject/myproject/.git/

enovo@LAPTOP-80NSFJM5 MINGW64 ~/Myproject/myproject (master)
$ touch file1.txt

enovo@LAPTOP-80NSFJM5 MINGW64 ~/Myproject/myproject (master)
$ git add file1.txt

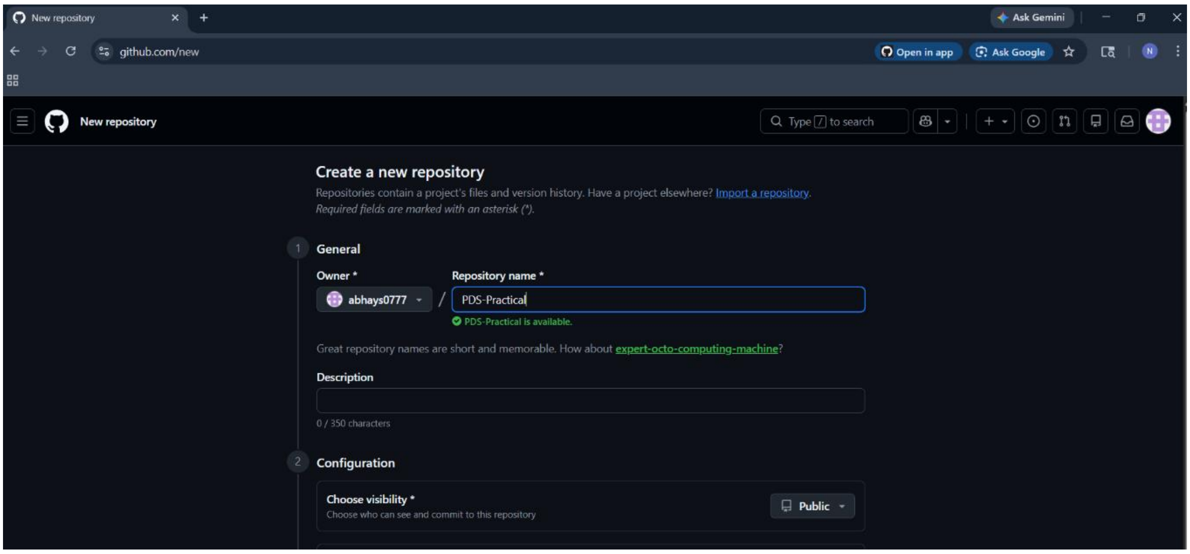
enovo@LAPTOP-80NSFJM5 MINGW64 ~/Myproject/myproject (master)
$ git add .

enovo@LAPTOP-80NSFJM5 MINGW64 ~/Myproject/myproject (master)
$ git commit -m "First commit"
[master (root-commit) 351da20] First commit
 1 file changed, 0 insertions(+), 0 deletions(-)
 create mode 100644 file1.txt

enovo@LAPTOP-80NSFJM5 MINGW64 ~/Myproject/myproject (master)
$ git remote add origin http://github.com/abhays0777/PDS-Practicals.git

enovo@LAPTOP-80NSFJM5 MINGW64 ~/Myproject/myproject (master)
$ git push -u origin master
info: please complete authentication in your browser...
warning: redirecting to https://github.com/abhays0777/PDS-Practicals.git/
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Writing objects: 100% (3/3), 211 bytes | 105.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
remote:
remote: Create a pull request for 'master' on GitHub by visiting:
remote:      https://github.com/abhays0777/PDS-Practicals/pull/new/master
remote:
To http://github.com/abhays0777/PDS-Practicals.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

enovo@LAPTOP-80NSFJM5 MINGW64 ~/Myproject/myproject (master)
$
```

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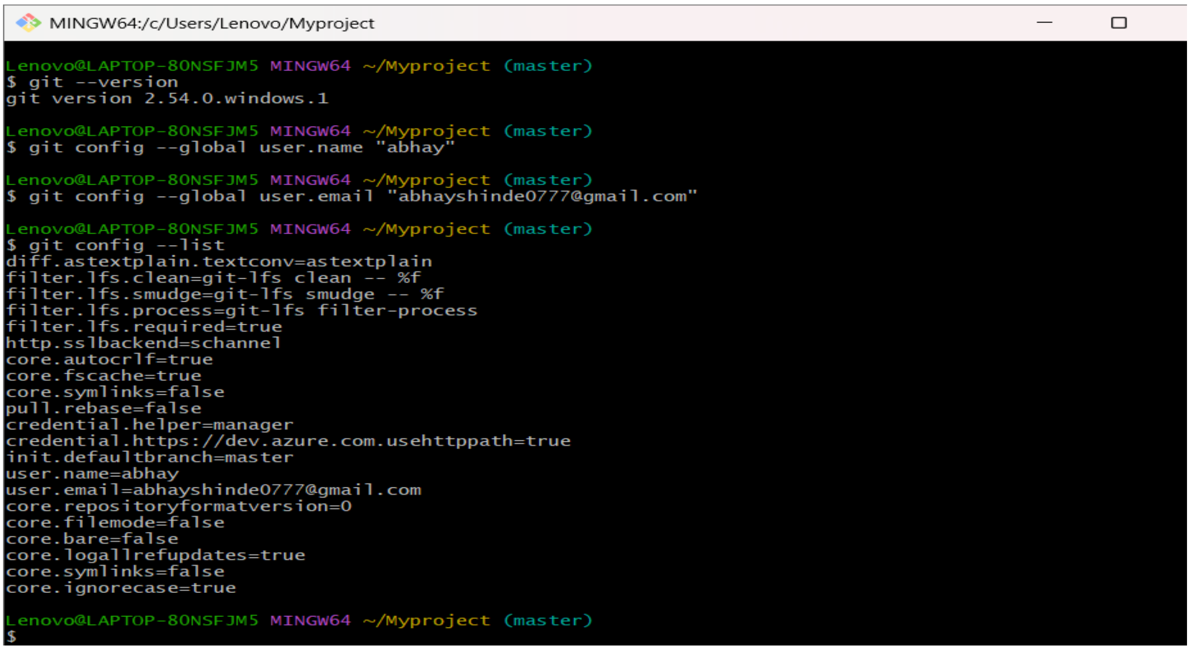
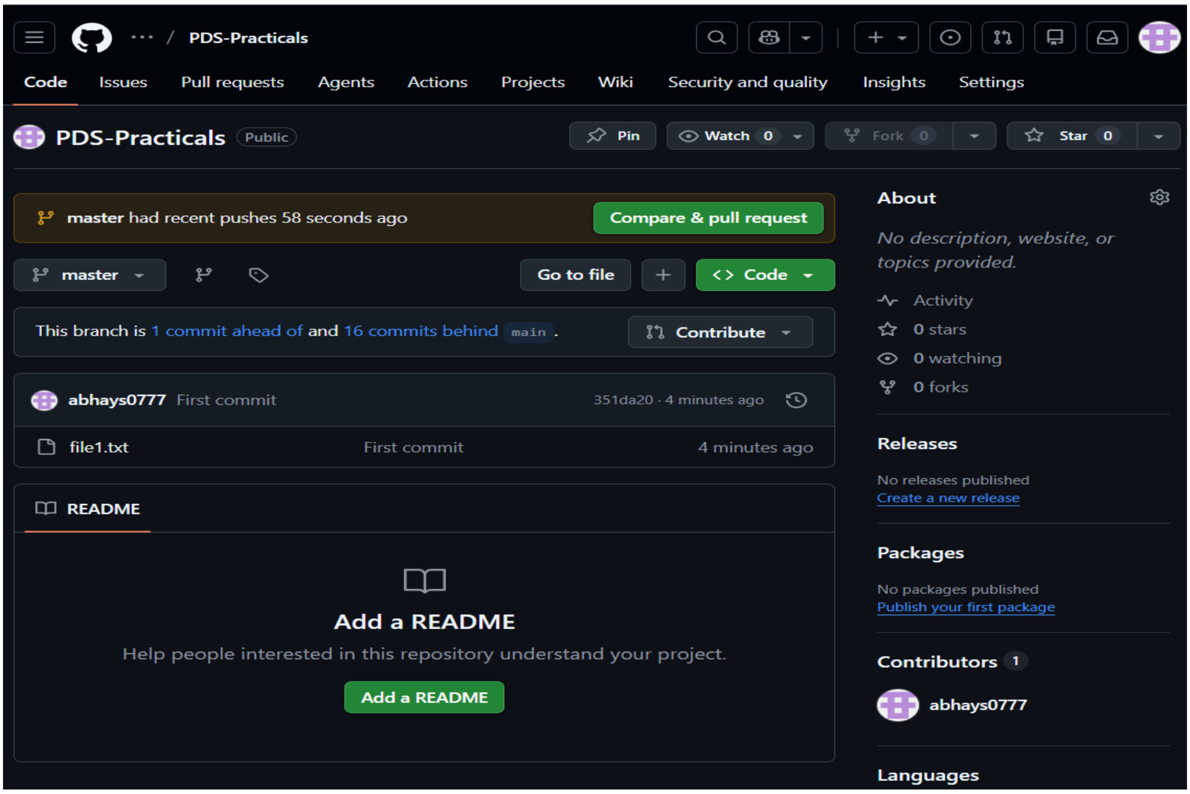
To http://github.com/abhays0777/PDS-Practicals.git
 * [new branch]      master -> master
branch 'master' set up to track 'origin/master'.

Lenovo@LAPTOP-80NSFJM5 MINGW64 ~/Myproject/myproject (master)
$
```


MINGW64:/c/Users/Lenovo/Myproject
Lenovo@LAPTOP-8ONSFJM5 MINGW64 ~/Myproject (master)
$ git --version
git version 2.54.0.windows.1
Lenovo@LAPTOP-8ONSFJM5 MINGW64 ~/Myproject (master)
$ git config --global user.name "abhay"
Lenovo@LAPTOP-8ONSFJM5 MINGW64 ~/Myproject (master)
$ git config --global user.email "abhayshinde0777@gmail.com"
Lenovo@LAPTOP-8ONSFJM5 MINGW64 ~/Myproject (master)
$ git config --list
diff.astextplain.textconv=astextplain
filter.lfs.clean=git-lfs clean -- %f
filter.lfs.smudge=git-lfs smudge -- %f
filter.lfs.process=git-lfs filter-process
filter.lfs.required=true
http.sslbackend=schannel
core.autocrlf=true
core.fscache=true
core.symlinks=false
pull.rebase=false
credential.helper=manager
credential.https://dev.azure.com.usehttppath=true
init.defaultbranch=master
user.name=abhay
user.email=abhayshinde0777@gmail.com
core.repositoryformatversion=0
core.filemode=false
core.bare=false
core.logallrefupdates=true
core.symlinks=false
core.ignorecase=true
Lenovo@LAPTOP-8ONSFJM5 MINGW64 ~/Myproject (master)
$


PDS-Practicals
Code Issues Pull requests Agents Actions Projects Wiki Security and quality Insights Settings
PDS-Practicals Public
Pin Watch 0 Fork 0 Star 0
master had recent pushes 58 seconds ago
Compare & pull request
master
Go to file
Code
This branch is 1 commit ahead of and 16 commits behind main .
Contribute
abhays0777 First commit
351da20 · 4 minutes ago
file1.txt First commit 4 minutes ago
README
Add a README
Help people interested in this repository understand your project.
Add a README
About
No description, website, or topics provided.
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
Contributors 1
abhays0777
Languages